

Class Test I
CITS2200 Data Structures and Algorithms
March 28, 2024

- Total marks for this paper is 10. Please answer by filling the appropriate circle for each question in your answer sheet.
- You should carefully fill your student number in the answer sheet.
- Please return only the answer sheet, **do not** return the question paper.

Q1. Which of the following statements is true about $O(n \log n)$ (a)

(a) $O(n \log n)$ is $O(n^2)$; (b) $O(n \log n)$ is $O(n)$; (c) $O(n \log n)$ is $O(\log n)$;
(d) none of the above.

Q2. The worst-case complexity of the Merge Sort algorithm is best expressed as: (c)

(a) $O(n^2)$; (b) $O(n)$; (c) $O(n \log n)$; (d) none of the above.

Q3. The worst-case complexity of merging two sorted arrays of lengths n_1 and n_2 ($n_1 + n_2 = n$) is best stated as: (b)

(a) $O(n^2)$; (b) $O(n)$; (c) $O(n \log n)$; (d) none of the above.

Q4. You have written a Python program for **Insertion Sort** for sorting an array of numbers in ascending order. The algorithm is given an input array that is already sorted in ascending order. Which of the following describes the best time complexity of your program on this input? (a)

(a) $O(n)$; (b) $O(n^2)$; (c) $O(n \log n)$; (d) none of the above.

Q5. Which one of the following states the complexity of binary search on a sorted array of length n ? (b)

(a) $O(n \log n)$; (b) $O(\log n)$; (c) $O(n^2)$; (d) $O(\log^2 n)$.

Q6. Which of the following statements about the references in a referential array in Python is true?

(a) Each reference takes an equal amount of space; (b) Different references take different amount of space; (c) References and the objects that they refer to take the same amount of space; (d) none of the above. (a)

Q7. Which of the following codes is correct for recursively computing the factorial of an integer?

(a) `def factorial(n):
 return n*factorial(n-1)`

(b) `def factorial(n):
 if n==0
 return 1
 else:
 return n*factorial(n-1)` (b)

(c) `def factorial(n):
 return factorial(n-1)`

(d) `def factorial(n):
 if n==0
 return 1
 else return n*factorial(n)`

Q8. You are given a sorted array A of length n , and another sorted array B of length k . k is much smaller than n , for example, k may be 100 and n may be 1 million. Your task is to find whether each element in the array B is present in the array A . What is the complexity of the best algorithm for this task? (b)

(a) $O(n \log k)$; (b) $O(k \log n)$; (c) $O(k^2)$; (d) $O(n^2)$.

Q9. There are three parts in an algorithm with complexities $O(n^2)$, $O(n \log n)$ and $O(n)$. What is the overall complexity of the algorithm? (b)

(a) $O(n \log n)$; (b) $O(n^2)$; (c) $O(n^4 \log n)$; (d) none of the above.

Q10. Given an unsorted array of n numbers, what is the complexity of the best algorithm for finding the maximum number in the array? (a)

(a) $O(n)$; (b) $O(n \log n)$; (c) $O(n^2)$; (d) $O(n^2 \log n)$.

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